

12 (long)

Figure 9 shows a simplified model which may be used to analyse the vibrations of a vehicle. The mass of the wheel is m and the mass of the vehicle it supports is $3m$. The masses are connected by a spring of stiffness k , accounting for the coil spring of the suspension. The mass m is contacting the road surface via a spring of stiffness $2k$ modelling the elasticity of the tyre. The vertical displacements of the masses $3m$ and m are y_1 and y_2 , respectively, measured upwards from the equilibrium position. The input excitation from road irregularities is x .

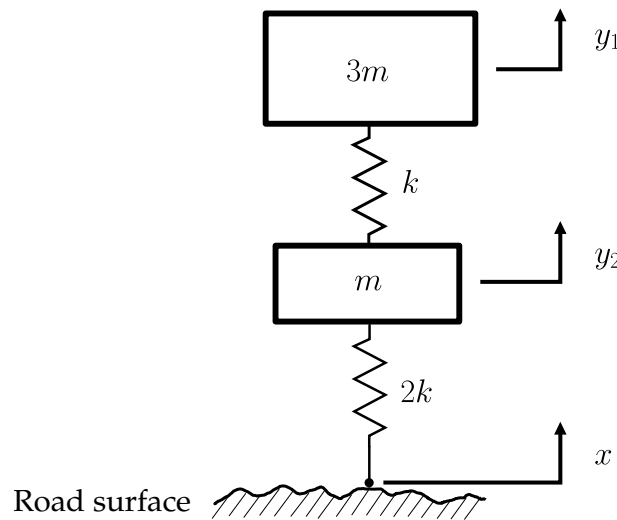


Fig. 9

- (a) Find mass and stiffness matrices for the system. Use these to find the natural frequencies and mode shapes of the system and sketch the mode shapes. [12]
- (b) For $x = X \cos(\omega t)$, the vehicle and wheel displacements will have the form $y_1 = Y_1 \cos(\omega t)$ and $y_2 = Y_2 \cos(\omega t)$, respectively. Sketch Y_1/X and Y_2/X as a function of ω . Label all salient values on both axes. [12]
- (c) Sketch $|Y_1/X|$ and $|Y_2/X|$ on a new figure, and show qualitatively the effect of damping in the tyre and suspension on these curves. [6]

END OF PAPER